

Math Thermo
class # 25
4/28/2026

Onsager - Casimir Relations

Theorem (25.1): Suppose that

$$(25.1) \quad \sigma_s = \sum_{i=1}^n J_i^\alpha X_i^\alpha + \sum_{i=1}^m J_i^\beta X_i^\beta,$$

where J_i^α, J_i^β are fluxes and X_i^α, X_i^β are even (α) and odd (β) thermodynamic forces. Assume that

$$(25.2) \quad J_i^\alpha = \sum_{j=1}^n L_{ij}^{\alpha\alpha} X_j^\alpha + \sum_{j=1}^m L_{ij}^{\alpha\beta} X_j^\beta, \quad 1 \leq i \leq n,$$

$$(25.3) \quad J_i^\beta = \sum_{j=1}^n L_{ij}^{\beta\alpha} X_j^\alpha + \sum_{j=1}^m L_{ij}^{\beta\beta} X_j^\beta, \quad 1 \leq i \leq m,$$

where

$$(25.4) \quad L_{ij}^{\alpha,\alpha} = L_{ji}^{\alpha,\alpha}, \quad 1 \leq i, j \leq n,$$

$$(25.5) \quad L_{ij}^{\beta,\beta} = L_{ji}^{\beta,\beta}, \quad 1 \leq i, j \leq m,$$

$$(25.6) \quad L_{ij}^{\alpha,\beta} = -L_{ji}^{\beta,\alpha}, \quad 1 \leq i \leq n, \quad 1 \leq j \leq m.$$

Set

$$L := \begin{bmatrix} L^{\alpha,\alpha} & L^{\alpha,\beta} \\ L^{\beta,\alpha} & L^{\beta,\beta} \end{bmatrix}, \quad \vec{J} := \begin{bmatrix} \vec{J}^\alpha \\ \vec{J}^\beta \end{bmatrix}, \quad \vec{X} := \begin{bmatrix} \vec{X}^\alpha \\ \vec{X}^\beta \end{bmatrix}.$$

Then

$$(25.7) \quad \sigma_s = X^T L_s X = \vec{X}^{\alpha T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\beta T} L^{\beta,\beta} \vec{X}^\beta,$$

where

$$L_S = \frac{1}{2}(L + L^T) = \begin{bmatrix} L^{\alpha, \alpha} & 0 \\ 0 & L^{\beta, \beta} \end{bmatrix}.$$

Proof: Clearly,

$$\sigma_S = \vec{J}^T \vec{X}$$

and

$$\vec{J} = L \vec{X}.$$

In particular,

$$\vec{J} = L \vec{X} = \begin{bmatrix} L^{\alpha, \alpha} \vec{X}^\alpha + L^{\alpha, \beta} \vec{X}^\beta \\ L^{\beta, \alpha} \vec{X}^\alpha + L^{\beta, \beta} \vec{X}^\beta \end{bmatrix}.$$

Thus, since $L^{\alpha, \alpha T} = L^{\alpha, \alpha}$ and $L^{\beta, \beta T} = L^{\beta, \beta}$,

$$\vec{J}^T = \left[\vec{X}^{\alpha T} L^{\alpha, \alpha} + \vec{X}^{\beta T} L^{\alpha, \beta T}, \vec{X}^{\alpha T} L^{\beta, \alpha T} + \vec{X}^{\beta T} L^{\beta, \beta} \right]$$

and

$$\sigma_S = \vec{J}^T \vec{X}$$

$$\begin{aligned} &= \vec{X}^{\alpha T} L^{\alpha, \alpha} \vec{X}^\alpha + \vec{X}^{\beta T} L^{\alpha, \beta T} \vec{X}^\alpha \\ &\quad + \vec{X}^{\alpha T} L^{\beta, \alpha T} \vec{X}^\beta + \vec{X}^{\beta T} L^{\beta, \beta} \vec{X}^\beta \end{aligned}$$

$$\begin{aligned} &= \vec{X}^{\alpha T} L^{\alpha, \alpha} \vec{X}^\alpha + \vec{X}^{\alpha T} L^{\alpha, \beta} \vec{X}^\beta \\ &\quad + \vec{X}^{\alpha T} L^{\beta, \alpha} \vec{X}^\beta + \vec{X}^{\beta T} L^{\beta, \beta} \vec{X}^\beta \end{aligned}$$

$$\begin{aligned}
&= \vec{X}^\alpha{}^\top L^{\alpha, \alpha} \vec{X}^\alpha + \vec{X}^\alpha{}^\top L^{\alpha, \beta} \vec{X}^\beta \\
&\quad - \vec{X}^\alpha{}^\top L^{\alpha, \beta} \vec{X}^\beta + \vec{X}^\beta{}^\top L^{\beta, \beta} \vec{X}^\beta \\
&= \vec{X}^\alpha{}^\top L^{\alpha, \alpha} \vec{X}^\alpha + \vec{X}^\beta{}^\top L^{\beta, \beta} \vec{X}^\beta,
\end{aligned}$$

Since

$$L^{\beta, \alpha}{}^\top = -L^{\alpha, \beta}. \quad //$$

The relations (25.4) - (25.6) are called Onsager - Casimir reciprocal relations.

Corollary (25.2): If $L^{\alpha, \alpha}$ and $L^{\beta, \beta}$ are positive semi-definite, then $\sigma_s \geq 0$.

Thus, we now have a more general framework for choosing thermodynamic fluxes to ensure that the entropy production term is always non-negative, $\sigma_s \geq 0$. We will use this in the next examples.

The only curious point to address is the separation of thermodynamic forces into even and odd variables, whatever this means. We will talk about this later.

Thermal Compatibility

Let us switch gears for the moment and address the curious property of thermal compatibility that we introduced in the last example.

let us suppose that

$$(25.8) \quad p_s = \hat{s} = \hat{s}(T, \phi_1, \dots, \phi_r)$$

$$(25.9) \quad p_u = \hat{u} = \hat{u}(T, \phi_1, \dots, \phi_r)$$

$$(25.10) \quad p_f = \hat{f} = \hat{f}(T, \phi_1, \dots, \phi_r),$$

with

$$(25.11) \quad \hat{f} = \hat{u} - T\hat{s}.$$

Assume that the thermal compatibility relation holds:

$$(25.12) \quad \frac{\partial \hat{s}}{\partial T} = \frac{1}{T} \frac{\partial \hat{u}}{\partial T}.$$

Consider the following calculation:

$$\begin{aligned} \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) &= \frac{\partial \hat{u}}{\partial T} \frac{\partial T}{\partial t} + \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \frac{\partial \phi_i}{\partial t} \\ &\quad + \nabla \hat{u} \cdot \vec{v} + \hat{u} \nabla \cdot \vec{v} \\ &= \frac{\partial \hat{u}}{\partial T} \frac{\partial T}{\partial t} + \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \frac{\partial \phi_i}{\partial t} \\ &\quad + \frac{\partial \hat{u}}{\partial T} \nabla T \cdot \vec{v} + \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \nabla \phi_i \cdot \vec{v} \\ &\quad + \hat{u} \nabla \cdot \vec{v} \end{aligned}$$

Thus,

$$(25.13) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = \frac{\partial \hat{u}}{\partial T} \frac{DT}{Dt} + \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \frac{D\phi_i}{Dt} + \hat{u} (\nabla \cdot \vec{v}).$$

likewise,

$$(25.14) \quad \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = \frac{\partial \hat{s}}{\partial T} \frac{DT}{Dt} + \sum_{i=1}^r \frac{\partial \hat{s}}{\partial \phi_i} \frac{D\phi_i}{Dt} + \hat{s} (\nabla \cdot \vec{v}).$$

Using thermal compatibility,

$$(25.15) \quad T \left(\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) \right) = \frac{\partial \hat{u}}{\partial T} \frac{DT}{Dt} + T \sum_{i=1}^r \frac{\partial \hat{s}}{\partial \phi_i} \frac{D\phi_i}{Dt} + T \hat{s} \nabla \cdot \vec{v}.$$

For simplicity, in this calculation, let us assume that the potential energies are negligible and there are no reactions. Thus,

$$(25.16) \quad \rho \frac{D\phi_i}{Dt} = -\nabla \cdot \vec{J}_i^D,$$

$$(25.17) \quad \rho \frac{D\vec{v}}{Dt} + \nabla P = -\nabla \cdot \Pi,$$

$$(25.18) \quad \frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_U^D - \Pi : \nabla \vec{v}.$$

In this case, the heat equation becomes

$$(25.19) \quad \frac{\partial \hat{u}}{\partial T} \frac{DT}{Dt} = -\nabla \cdot \vec{J}_U^D - \Pi : \nabla \vec{v} - \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \frac{D\phi_i}{Dt} - \hat{u} (\nabla \cdot \vec{v}).$$

Combining (25.19) and (25.15), we have

$$\begin{aligned}
T \left(\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) \right) &= -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v} \\
&\quad - \sum_{i=1}^r \frac{\partial \hat{u}}{\partial \phi_i} \frac{D \phi_i}{Dt} + T \sum_{i=1}^r \frac{\partial \hat{s}}{\partial \phi_i} \frac{D \phi_i}{Dt} \\
&\quad - \hat{u}(\nabla \cdot \vec{v}) + T \hat{s}(\nabla \cdot \vec{v}). \\
&= -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v} \\
&\quad - \sum_{i=1}^r \left(\frac{\partial \hat{u}}{\partial \phi_i} - T \frac{\partial \hat{s}}{\partial \phi_i} \right) \frac{D \phi_i}{Dt} \\
&\quad - \hat{f}(\nabla \cdot \vec{v}) \\
&= -\nabla \cdot \vec{J}_U^0 - \Pi : \nabla \vec{v} \\
&\quad - \sum_{i=1}^r \frac{\partial \hat{f}}{\partial \phi_i} \frac{D \phi_i}{Dt} - \hat{f}(\nabla \cdot \vec{v}).
\end{aligned}$$

Thus,

$$\begin{aligned}
\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) &= -\frac{1}{T} \nabla \cdot \vec{J}_U^0 - \frac{\Pi : \nabla \vec{v}}{T} \\
&\quad - \frac{1}{T} \sum_{i=1}^r \frac{\partial \hat{f}}{\partial \phi_i} \frac{D \phi_i}{Dt} \\
&\quad - \frac{\hat{f}}{T} (\nabla \cdot \vec{v}).
\end{aligned}$$

Using (25.16), we, therefore, have

$$\begin{aligned}
\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) &= -\frac{1}{T} \nabla \cdot \vec{J}_U^0 - \frac{\Pi : \nabla \vec{v}}{T} \\
&\quad + \frac{1}{\rho T} \sum_{i=1}^r \frac{\partial \hat{f}}{\partial \phi_i} \nabla \cdot \vec{J}_i^0 \\
&\quad - \frac{\hat{f}}{T} \nabla \cdot \vec{v}.
\end{aligned}$$

(25.20)

Now, since

$$\sum_{i=1}^r \vec{J}_i^D = 0$$

$$\frac{1}{\rho T} \sum_{i=1}^r \frac{\partial \hat{f}}{\partial \phi_i} \nabla \cdot \vec{J}_i^D = 0$$

Hence,

$$(25.21) \quad \begin{aligned} \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) &= -\frac{1}{T} \nabla \cdot \vec{J}_U^D - \frac{\Pi : \nabla \vec{v}}{T} \\ &+ \frac{1}{\rho T} \sum_{i=2}^r \left(\frac{\partial \hat{f}}{\partial \phi_i} - \frac{\partial \hat{f}}{\partial \phi_1} \right) \nabla \cdot \vec{J}_i^D \\ &- \frac{\hat{f}}{T} \nabla \cdot \vec{v}. \end{aligned}$$

We make the identification

$$\mu_i := \frac{1}{\rho} \frac{\partial \hat{f}}{\partial \phi_i}, \quad i=1, \dots, r.$$

As before, we set

$$\tilde{\mu}_i := \mu_i - \mu_1, \quad i=2, \dots, r,$$

which is called the exchange chemical potential.
Thus,

$$\begin{aligned} \rho \frac{Ds}{Dt} &= \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) \\ &= -\frac{1}{T} \nabla \cdot \vec{J}_U^D - \frac{\Pi : \nabla \vec{v}}{T} \\ &+ \sum_{i=2}^r \frac{\tilde{\mu}_i}{T} \nabla \cdot \vec{J}_i^D - \frac{\hat{f}}{T} (\nabla \cdot \vec{v}). \end{aligned}$$

Recall

$$\rho \frac{Ds}{Dt} = \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v})$$

always holds if we define $p_s = \hat{s}$ and ρ satisfies the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0,$$

where $\vec{v}$ is the barycentric velocity.

As usual, we use the identities

$$\frac{1}{T} \nabla \cdot \vec{J}_v^D = \nabla \cdot \left(\frac{1}{T} \vec{J}_v^D \right) - \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^D$$

and

$$\frac{\tilde{\mu}_i}{T} \nabla \cdot \vec{J}_i^D = \nabla \cdot \left(\frac{\tilde{\mu}_i}{T} \vec{J}_i^D \right) - \nabla \left(\frac{\tilde{\mu}_i}{T} \right) \cdot \vec{J}_i^D,$$

obtaining

$$(25.22) \quad \begin{aligned} \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = & - \nabla \cdot \left(\frac{1}{T} \vec{J}_v^D \right) + \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^D \\ & + \sum_{i=2}^r \nabla \cdot \left(\frac{\tilde{\mu}_i}{T} \vec{J}_i^D \right) - \frac{\pi : \nabla \vec{v}}{T} \\ & - \sum_{i=2}^r \nabla \left(\frac{\tilde{\mu}_i}{T} \right) \cdot \vec{J}_i^D - \frac{\hat{f}}{T} (\nabla \cdot \vec{v}). \end{aligned}$$

Hence,

$$(25.23) \quad \rho \frac{Ds}{Dt} = \frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = - \nabla \cdot \vec{J}_s^D + \sigma_s,$$

where

$$(25.24) \quad \vec{J}_s^D = \frac{1}{T} \left(\vec{J}_v^D - \sum_{i=2}^r \tilde{\mu}_i \vec{J}_i^D \right)$$

and

$$(25.25) \quad \begin{aligned} \sigma_s = & \nabla \left(\frac{1}{T} \right) \cdot \vec{J}_v^D - \sum_{i=2}^r \nabla \left(\frac{\tilde{\mu}_i}{T} \right) \cdot \vec{J}_i^D \\ & - \frac{1}{T} \nabla \vec{v} : (\pi + \hat{f} \mathbf{I}_3). \end{aligned}$$

In short, when we use the thermal compatibility conditions, we obtain the same basic forms as before.

To see this, compare the results of (25.24) and (25.25) with what was obtained in lectures 21 and 22.

Curie's Principle

let us again recall the entropy production term:

$$(25.26) \quad \sigma_s = T \vec{J}_s^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{\rho \nabla \cdot \vec{v}}{3T} - \frac{\tau : \Pi}{T} \\ - \frac{1}{T} \sum_{i=2}^r \vec{J}_i^D \cdot \nabla \tilde{\mu}_i^e - \frac{1}{T} \sum_{k=1}^K A_k \tilde{\mathcal{F}}_k.$$

The particular form here is taken from (22.13). Recall that

$$T \vec{J}_s^D = \vec{J}_0^D - \sum_{i=2}^r \tilde{\mu}_i^e \vec{J}_i^D$$

A_k is the k^{th} chemical affinity,

$$A_k := \sum_{i=1}^r \nu_{i,k} \mu_i, \quad k=1, \dots, K.$$

$\tilde{\mu}_i^e$ is the exchange extended chemical potential:

$$\tilde{\mu}_i^e = \mu_i^e - \mu_1^e, \quad \mu_i^e = \mu_i + \psi_i.$$

τ is the shear rate tensor:

$$\tau := \frac{1}{2} (\nabla \vec{v} + (\nabla \vec{v})^T) - \frac{1}{3} \nabla \cdot \vec{v} \mathbb{I}_3.$$

Finally recall that the Cauchy stress tensor, $\mathcal{S}$, which appears in the momentum balance equation

$$\frac{\partial(\rho \vec{v})}{\partial t} + \nabla \cdot (\rho \vec{v} \otimes \vec{v} + \mathcal{S}) = - \sum_{i=1}^r \rho_i \nabla \psi_i,$$

has been decomposed as

$$\mathcal{S} = P \mathbf{I}_3 + p \mathbf{I}_3 + \hat{\Pi},$$

where P is the (scalar) hydrostatic pressure, p is the (scalar) non-hydrostatic pressure, and $\hat{\Pi}$ is the deviatoric (symmetric and trace-free) part of

$$\Pi = \mathcal{S} - P \mathbf{I}_3.$$

Curie's Principle: Fluxes and forces of different tensorial characters do not couple.

This has an important implication for the Onsager-Cosserat reciprocal relations. In particular, we write

$$(25.27) \quad \sigma_S = \sigma_{S,0} + \sigma_{S,1} + \sigma_{S,2},$$

where

$$(25.28) \quad \sigma_{S,0} := - \frac{p \nabla \cdot \vec{v}}{3T} - \frac{1}{T} \sum_{k=1}^K A_k \vec{F}_k, \quad (\text{order } 0)$$

$$(25.29) \quad \sigma_{S,1} := T \vec{J}_S^D \cdot \nabla \left(\frac{1}{T} \right) - \frac{1}{T} \sum_{i=2}^r \vec{J}_i^D \cdot \nabla \tilde{\mu}_i^e, \quad (\text{order } 1)$$

and

$$(25.30) \quad \sigma_{s,2} := - \frac{\gamma : \dot{\Pi}}{\dot{T}} \quad (\text{order } 2)$$

The forces (according to their tensor orders) are

$$\begin{aligned} & - \frac{\nabla \cdot \vec{v}}{3T}, \quad - \frac{A_k}{T}, \quad k=1, \dots, K, \quad (\text{order } 0) \\ & T \nabla \left(\frac{1}{T} \right), \quad - \frac{1}{T} \nabla \tilde{\mu}_i^e, \quad i=2, \dots, r, \quad (\text{order } 1) \\ & - \frac{\gamma}{T}. \quad (\text{order } 2) \end{aligned}$$

These are the known quantities. The unknowns, to be determined by constitutive relations, are following thermodynamic fluxes:

$$\begin{aligned} & p, \quad \tilde{\sigma}_k, \quad k=1, \dots, K, \quad (\text{order } 0) \\ & \vec{J}_s^D, \quad \vec{J}_i^D, \quad i=2, \dots, r, \quad (\text{order } 1) \\ & \dot{\Pi}. \quad (\text{order } 2) \end{aligned}$$

Gies' Principle indicates that

$$(25.31) \quad \begin{aligned} p &= - l_{1,1}^{v,v} \frac{\nabla \cdot \vec{v}}{3T} - \sum_{j=1}^K l_{1,j}^{v,a} \frac{A_j}{T}, \\ \tilde{\sigma}_i &= - l_{i,1}^{a,v} \frac{\nabla \cdot \vec{v}}{3T} - \sum_{j=1}^K l_{i,j}^{a,a} \frac{A_j}{T}, \quad i=1, \dots, K, \end{aligned} \quad (\text{order } 0)$$

$$(25.32) \quad \begin{aligned} \vec{J}_s^D &= L_{1,1}^{q,q} T \nabla \left(\frac{1}{T} \right) - \sum_{j=2}^r L_{1,j}^{q,\mu} \frac{1}{T} \nabla \tilde{\mu}_j^e, \\ \vec{J}_i^D &= L_{i,1}^{\mu,q} T \nabla \left(\frac{1}{T} \right) - \sum_{j=2}^r L_{i,j}^{\mu,\mu} \frac{1}{T} \nabla \tilde{\mu}_j^e, \quad i=2, \dots, r, \end{aligned} \quad (\text{order } 1)$$

$$(25.33) \quad \dot{\Pi} = - L_{1,1}^{\gamma,\gamma} \frac{1}{T} \gamma. \quad (\text{order } 2)$$

The theory of Onsager and Casimir indicates that $\nabla \cdot \vec{v}$ and A_k have different parity. Thus, the reciprocal relations are

$$-l_{1,j}^{v,a} = l_{j,1}^{a,v}, \quad j=1, \dots, K; \quad l_{ij}^{a,a} = l_{ji}^{a,a}, \quad i,j=1, \dots, K.$$

On the other hand, $T \nabla(\frac{1}{T})$ and $\frac{1}{T} \nabla \tilde{\mu}_i^e$ have the same parity. Thus,

$$L_{1,j}^{q,\mu} = L_{j,1}^{\mu,q}, \quad j=2, \dots, r,$$

$$L_{i,j}^{\mu,\mu} = L_{j,i}^{\mu,\mu}, \quad i,j=2, \dots, r.$$

For non-negative entropy production, we must demand that the following matrices are SPD:

$$\mathcal{L} = \begin{bmatrix} l_{1,1}^{v,v} & 0 & \dots & 0 \\ 0 & l_{1,1}^{a,a} & \dots & l_{1,K}^{a,a} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & l_{K,1}^{a,a} & \dots & l_{K,K}^{a,a} \end{bmatrix} \in \mathbb{R}^{(K+1) \times (K+1)},$$

$$\mathcal{L} = \begin{bmatrix} L_{1,1}^{q,q} & L_{1,2}^{q,\mu} & \dots & L_{1,r}^{q,\mu} \\ L_{2,1}^{\mu,q} & L_{2,2}^{\mu,\mu} & \dots & L_{2,r}^{\mu,\mu} \\ \vdots & \vdots & \ddots & \vdots \\ L_{r,1}^{\mu,q} & L_{r,2}^{\mu,\mu} & \dots & L_{r,r}^{\mu,\mu} \end{bmatrix} \in \mathbb{R}^{r \times r},$$

$$\mathcal{L} = [\mathcal{L}_{1,1}^{r,r}] \in \mathbb{R}^{1 \times 1}.$$